

UNIT IV

17. Solve

(a) $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$

(b) $\frac{dy}{dx} + \frac{\cos x}{\sin x} y = \cos x$.

Or

18. (a) Solve $p^2x - 2py - x = 0$.

(b) Solve $xy = (1+p)x + p^2$.

UNIT V

19. Solve $(D^2 - 4D + 4)y = 8x^2e^{2x} \sin 2x$.

Or

20. Solve $(x^2D^2 - xD + 4)y = x^2 \sin(\log x)$.

5781001

B.Tech. DEGREE EXAMINATION, MAY 2019.

First Semester

Common to All branches

MATHEMATICS — I

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Define radius of curvature.
2. Prove that $\lceil (n+1) = n! \rceil$ where n is a positive integer.

3. If $u = xy + yz + zx$ where $x = e^t$, $y = e^{-t}$ and $z = \frac{1}{t}$ find $\frac{du}{dt}$.

4. If $u = 2xy$, $v = x^2 - y^2$, $x = r \cos \theta$, $y = r \sin \theta$ compute $\frac{\partial(u,v)}{\partial(r,\theta)}$.

5. Evaluate $\int_0^{2\pi} \int_0^\pi \int_0^a r^4 \sin \phi \, dr \, d\phi \, d\theta$.

6. Define volume integral.

7. Solve the equation $\left(\frac{dy}{dx}\right)^2 - 8\frac{dy}{dx} + 15 = 0$.

8. Solve $(D^4 - 2D^3 + D^2)y = 0$.

9. Solve $y = 2px + yp^2$ by converting it as a Clairaut's equation.

10. Solve $(e^x + 1) \cos x \, dx + e^x \sin x \, dy$ by variable separable method.

PART B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each Unit.

UNIT I

11. Find the centre of curvature at $\theta = \pi/2$ on the curve $x = 2 \cos t + \cos 2t$, $y = 2 \sin t + \sin 2t$.

Or

12. Show that $\beta(n, n) = \frac{\sqrt{\pi} \Gamma(n)}{2^{2n+1} \Gamma\left(n + \frac{1}{2}\right)}$.

$\beta = \frac{1}{\beta_2} = \frac{dy}{dx}$

2

5781001

UNIT II

13. (a) If $x^2 + y^2 + z^2 - 2xyz = 1$ show that

$$\frac{dx}{\sqrt{1-x^2}} + \frac{dy}{\sqrt{1-y^2}} + \frac{dz}{\sqrt{1-z^2}} = 0.$$

(b) Examine the extreme of

$$f(x, y) = x^2 + xy + y^2 + \frac{1}{x} + \frac{1}{y}.$$

Or

14. Using Taylor's series verify that

$$\log(1 + x + y) = (x + y) + \frac{1}{2}(x + y)^2 + \frac{1}{3}(x + y)^3 + \dots$$

UNIT III

15. By changing the order of integration evaluate the following integrals.

(a) $\int_0^a \int_0^a \frac{x}{\sqrt{x^2 + y^2}} \, dx \, dy$

(b) $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} \, dy \, dx.$

Or

16. Evaluate $\iiint_V (x + y + z) \, dx \, dy \, dz$ where V is the region of space inside the cylinder $x^2 + y^2 = a^2$ which is bounded by the planes $z = 0$ and $z = h$.

3

5781001

UNIT IV

17. (a) Show that

$$(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0.$$

- (b) Solve $(a^2 - 2xy - y^2)dx - (x + y)^2dy = 0$.

Or

18. (a) Solve $y = (x - a)p - p^2$.

- (b) Solve $y = px + \frac{a}{p}$.

UNIT V

19. Solve $(D^3 + D^2 + D + 1)y = 2x^3 + 3x^2 - 4x + 5$.

Or

20. Solve $(D^2 - 4D + 3)y = \sin 3x \cos 2x$.

6. Evaluate $\int_0^1 \int_0^2 \int_1^2 x^2 yz \, dx \, dy \, dz$.

7. Write down any rule for finding integrating factors.

8. Find the I.F of the equation $(1-x^2) \frac{dy}{dx} - xy = 1$.

9. Solve $(D^3 - 6D^2 + 11D - 6)Y = 0$.

10. Define the homogeneous linear equation.

SECTION B — (5 × 11 = 55 marks)

Answer ALL, choosing ONE full question from each Unit.

UNIT I

11. Show that the radius of curvature at any point of the cycloid $x = a(\theta + \sin \theta), y = a(1 - \cos \theta)$ is $4a \cos \frac{\theta}{2}$.

Or

12. Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

2

5781001

UNIT II

13. If $V = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$, then prove that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$.

Or

14. Find the volume of the greatest rectangular Parallelopiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

UNIT III

15. (a) Find the area of a plate in the form of a quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$.

(b) Show that the area between the parabolas $y^2 - 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$.

Or

16. Find the volume of the portion of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 = ax$.

3

5781001

UNIT IV

17. A hot body cools in air at a rate proportional to the difference between the temperature of the body and that of the surrounding air. If the air is maintained at 20°C and that of the body cools from 100°C to 75°C in 10 minutes, when will its temperature be 25°C ? What will be its temperature in half an hour since it started cooling from 100°C ? (11)

Or

18. Solve $p^2y + 2px - y = 0$. (11)

UNIT V

19. Solve $(D^2 + 1)y = x^2e^{2x} + x\cos x$. (11)

Or

20. Solve by method of variation of parameter $\frac{d^2y}{dx^2} + n^2y = \sec nx$. (11)

5781001

B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2018.

First Semester

(Common to all branches)

MATHEMATICS - I

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Define circle of curvature.
2. Define the Gamma function.
3. Find $\frac{du}{dt}$ if $u = x^3y^4$ where $x = t^3$ and $y = t^2$.
4. If $x = u(1-v)$, $y = uv$ verify that $\frac{\partial(x,y)}{\partial(u,v)} \times \frac{\partial(u,v)}{\partial(x,y)} = 1$.

5. Evaluate $\int_0^{\pi/2} \int_0^{\pi/2} (a \cos 2\theta + b \sin 2\theta) d\theta d\phi$.

6. Evaluate $\int_0^2 \int_0^2 xy(x+y) dx dy$.

7. Show that $\int_0^1 (5x^4 + 3x^2y^2 - 2xy^3) dx (2x^3y - 3x^2y^2 - 5y^4) dy = 0$ is an exact equation.

8. Solve $p^2 - 5p + 6 = 0$ where $p = \frac{dy}{dx}$.

9. Solve the equation $(D^2 + 2D - 24)y = 0$.

10. Find the particular integral of $(D^2 + D + 1)y = x^2$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each unit.

UNIT I

11. Find the evolute of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. (11)

Or

12. Derive the relationship between Beta and Gamma functions. (11)

2

5781001

UNIT II

13. (a) If $u = x^3 + y^3 + z^3 + 3xyz$ find

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}. \quad (4)$$

(b) If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$ show that $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin \theta$. (7)

Or

14. Find the extreme values of the function $u(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$. (11)

UNIT III

15. Change of order of integration and hence evaluate $\int_{-a}^a \int_0^{\sqrt{a^2 - y^2}} x dx dy$. (11)

Or

16. Find the volume enclosed by the cylinders $x^2 + y^2 = 2ax$ and $z^2 = 2ax$ using double integral. (11)

3

5781001

UNIT V

19. Solve $(D^2 - 3D + 2)y = 2\cos(2x + 3) + 2e^x$.

Or

20. Solve $(D^2 + a^2)y = \tan ax$ by the method of variation of parameters.

5781001

B.Tech. DEGREE EXAMINATION, JANUARY 2016.

First Semester

MATHEMATICS — I

(Common to all branches)

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Find the radius of curvature of the curve $y = 2x^3 - 3x + 2$ at (1, 1).
2. Define evolute and involute.
3. Find the stationary points of $(x, y) = x^2 - xy + y^2 - 2x + y$.
4. If $x = r \cos \theta$, $y = r \sin \theta$, find $\frac{(x, y)}{(r, \theta)}$.

5. Evaluate $\int_2^b \int_2^a \frac{dx dy}{xy}$.

6. Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ by changing the order of integration.

7. Solve $(y - px)(p - 1) = p$.

8. Solve $p^2 - 7p + 12 = 0$.

9. Solve $(D^2 + 3D + 2)y = 0$.

10. Eliminate y from the simultaneous equations

$$\frac{dx}{dt} + y = \sin t; \quad \frac{dy}{dt} + x = \cos t.$$

SECTION B — (5 × 11 = 55 marks)

Answer ALL Units, choosing ONE question from each Unit.

UNIT I

11. Find the radius of curvature at $(a, 0)$ on the curve $xy^2 = a^3 - x^3$.

Or

12. Prove that the evolute of the rectangular hyperbola $xy = c^2$ is $(x + y)^{2/3} - (x - y)^{2/3} = (4c)^{2/3}$.

2

5781001

UNIT II

13. Find the Taylor's series expansion of $f(x, y) = x^2 y + \sin y + e^x$ about the point $(1, \pi)$.

Or

14. Find the maximum and minimum values of $f(x, y) = 2(x^2 - y^2) - x^4 + y^4$.

UNIT III

15. Evaluate $\int_0^{1-x} \int_0^{1-x} xy dy dx$ by changing the order of integration.

Or

16. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} xyz dx dy dz$.

UNIT IV

17. Solve $p^2 + 2py \cot x = y^2$.

Or

18. Solve $y = 2px + p^2$.

3

5781001

UNIT III

15. Evaluate by changing to polars, the integral

$$\int_0^a \int_y^b \frac{x^2 dx dy}{\sqrt{x^2 + y^2}} \quad (11)$$

Or

16. Find the volume of that portion of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, which lies in the first octant. (11)

UNIT IV

17. (a) Solve $(x^4 e^x - 2mxy)dx + 2mx^2 y dy = 0$. (5)

(b) Solve $(x^2 - yx^2) \frac{dy}{dx} + (y^2 + x^2 y^2) = 0$. (6)

Or

18. Solve $p^3 - 4xyp + 8y^2 = 0$. (11)

UNIT V

19. Solve the simultaneous equation $\frac{dx}{dt} + 2x - 3y = t$, $\frac{dy}{dt} - 3x + 2y = e^{2t}$. (11)

Or

20. Solve by the method of variation of parameters, the equation $\frac{d^2 y}{dx^2} + 4y = 4 \tan 2x$. (11)

5781001

B.Tech. DEGREE EXAMINATION,
NOVEMBER 2016.

First Semester

MATHEMATICS - I

(Common to all branches)

(From 2013-14 batch onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

- Write down the formula for radius of curvature in Cartesian Coordinates.
- Write down the relationship between beta and gamma function.
- If $u = (x - y)^4 + (y - z)^4 + (z - x)^4$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
- In polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, show that $\frac{\partial(x, y)}{\partial(r, \theta)} = r$.

5. Evaluate $\int_0^a \int_0^b xy(x-y) dx dy$.
6. Write down the total volume of the solid of revolution.
7. Solve $y dx - x dy - 3x^2 y^2 e^{x^3} dx = 0$.
8. Solve $p^2 - 9p + 18 = 0$ where $p = \frac{dy}{dx}$.
9. Solve $(D^3 + D^2 - D - 1)y = 0$.
10. Find the P.I. of the differential equation $(D^2 - 2D + 1)y = e^x \sin x$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE questions from each Unit.

UNIT I

11. (a) Find the radius of curvature at the point $(a \cos^3 \theta, a \sin^3 \theta)$ on the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$.
- (b) Evaluate $\int_0^{\frac{\pi}{2}} \cos^9 \theta d\theta$ using Gamma function.

Or

2 5781001

12. Show that the centre of curvature at any point (x, y) on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\left(\frac{a^2 - b^2}{a^4} x^3, -\frac{a^2 - b^2}{b^4} y^3 \right)$ and hence show that the equation of the evolute is $(ax)^{\frac{2}{3}} (by)^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}}$. (11)

UNIT II

13. (a) If $u = \sin^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$. (5)
- (b) If $x = r \cos \theta, y = r \sin \theta$ find $\frac{\partial(x, y)}{\partial(r, \theta)}$ and $\frac{\partial(r, \theta)}{\partial(x, y)}$ then verify that $\frac{\partial(x, y)}{\partial(r, \theta)} \frac{\partial(r, \theta)}{\partial(x, y)} = 1$. (6)

Or

14. Expand $xy^2 + 2x - 3y$ in powers of $(x+2)$ and $(y-1)$ upto the degree terms. (11)

3

5781001

16. Find the volume of that portion of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ which lies in the first octant using triple integration.

UNIT IV

17. (a) Find the orthogonal trajectories on the conicoid $(x+y)z=1$ of the conics in which it is cut by the system of planes $x-y+z=k$, where k is parameter. (6)
 (b) Solve $(px-y)(py+x)=2p$. (5)
- Or
18. (a) Solve $p^2+2yp \cot x - y^2 = 0$. (5)
 (b) Solve $y=3px+6p^2y^2$. (6)

UNIT V

19. (a) Solve $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$. (5)
 (b) Solve $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = \sin(\log x)$. (6)
- Or
20. Solve the simultaneous equations and $\frac{d^2y}{dt^2} + y = te^{3t}$ and $\frac{d^2y}{dt^2} + y - 2x = \cos^2 t$.

5781001

B.Tech. DEGREE EXAMINATION, MAY 2017.

First Semester

MATHEMATICS — I

(Common for ALL branches)

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Define curvature and radius of curvature.
2. Prove that $B(m, n) = B(n, m)$.
3. If $x^y + y^x = a^k$ find $\frac{dy}{dx}$.
4. If $u = 2xy$, $v = x^2 - y^2$ while $x = r \cos \theta$, $y = r \sin \theta$ prove that $\frac{\partial(u, v)}{\partial(r, \theta)} = 4r^3$.
5. Evaluate $\int_{-1}^{2} \int_x^{x+2} dy dx$.

6. Change of the order of the integration

$$\int_0^{\infty} \int_y^{\infty} \frac{e^{-y}}{y} dy dx.$$

7. Find the general solution of $y = xp + \frac{\alpha}{p}$.
8. Define exact differential equation.
9. Explain second order linear differential equation.
10. Solve $(D^2 - 3D + 2)y = e^{5x}$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full question from each Unit.

UNIT I

11. (a) Show that $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \times \int_0^{\infty} \frac{dx}{\sqrt{1+x^4}} = \frac{\pi\sqrt{2}}{4}$. (5)
- (b) Find the centre of curvature and circle of curvature at $(\frac{a}{4}, \frac{a}{4})$ on $\sqrt{x} + \sqrt{y} = \sqrt{a}$. (6)

Or

12. (a) Obtain the equation of the evolute of the ellipse. (8)
- (b) Prove that the radius of curvature at any point of the cycloid
- $$x = a(\theta + \sin \theta), y = a(1 - \cos \theta) \text{ is } 4a \cos \frac{\theta}{2}. \quad (3)$$

UNIT II

13. (a) If $u = \log r$ where $r^2 = (x-a)^2 + (y-a)^2$ and $(x-a)$ and $(y-b)$ are not simultaneous zero, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. (6)
- (b) Expand $x^2 y + 3y - 2$ in power of $(x-1)$ and $(y+2)$ upto the third degree terms. (5)

Or

14. (a) If $x = e^r \sec \theta$, $y = e^r \tan \theta$ prove $\frac{\partial(x, y)}{\partial(r, \theta)} \frac{\partial(r, \theta)}{\partial(x, y)} = 1$. (5)
- (b) Find the volume of the greatest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. (6)

UNIT III

15. (a) Evaluate $\int_0^{3\sqrt{4-y}} \int_0^1 (x+y) dx dy$ by changing the order of integration. (8)
- (b) Change the order of integration in
- $$I = \int_{-a}^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) dx dy.$$

Or

UNIT IV

17. Solve $p^2 - p (\cos x + \sec x) + 1 = 0$. (11)

Or

18. Solve $x(1-x^2)\frac{dy}{dx} + (2x^3-1)y = x^3$. (11)

UNIT V

19. Solve by method of variation of parameter $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$. (11)

Or

20. Solve $\frac{dx}{dt} + y = \sin t$; $\frac{dy}{dt} + x = \cos t$ when $x=2$, $y=0$ at $t=0$. (11)

8

4

5781001

5781001

B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2017.

First Semester
MATHEMATICS — I

(Common for All branches)
(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. Write the Cartesian formula for radius of curvature. $\rightarrow$

2. Define gamma and beta function.

If $u = \frac{x}{z} + \frac{y}{z} + \frac{z}{x}$ then find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$. $\rightarrow 0$

4. If $x = r \cos \theta$, $y = r \sin \theta$ then find $\frac{\partial(x,y)}{\partial(r,\theta)}$. $\rightarrow 5$

5. Evaluate $\int_0^1 \int_0^1 (x^2 + y^2) dy dx$. $\rightarrow 2$

$$\begin{aligned} & \int_0^1 \left[\frac{x^3}{3} + \frac{y^3}{3} \right]_0^1 dx \\ &= \int_0^1 \left(\frac{x^3}{3} + \frac{1}{3} \right) dx \\ &= \left[\frac{x^4}{12} + \frac{x}{3} \right]_0^1 \\ &= \frac{1}{12} + \frac{1}{3} = \frac{5}{12} \end{aligned}$$

6. Change of the order of the integration

$$\int_0^{\infty} \int_0^{\infty} \frac{e^{-y}}{y} dy dx.$$

7. Find the general solution of $y = xp + \frac{\alpha}{p}$.
8. Define exact differential equation.
9. Explain second order linear differential equation.
10. Solve $(D^2 - 3D + 2)y = e^{5x}$.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full question from each Unit.

UNIT I

11. (a) Show that $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \times \int_0^{\infty} \frac{dx}{\sqrt{1+x^4}} = \frac{\pi\sqrt{2}}{4}$. (5)

- (b) Find the centre of curvature and circle of curvature at $(\frac{a}{4}, \frac{a}{4})$ on $\sqrt{x} + \sqrt{y} = \sqrt{a}$. (6)

Or

12. (a) Obtain the equation of the evolute of the ellipse. (8)

- (b) Prove that the radius of curvature at any point of the cycloid

$$x = a(\theta + \sin \theta), y = a(1 - \cos \theta) \text{ is } 4a \cos \frac{\theta}{2}. (3)$$

UNIT II

13. (a) If $u = \log r$ where $r^2 = (x-a)^2 + (y-a)^2$ and $(x-a)$ and $(y-b)$ are not simultaneous zero, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. (6)

- (b) Expand $x^2y + 3y - 2$ in power of $(x-1)$ and $(y+2)$ upto the third degree terms. (5)

Or

14. (a) If $x = e^r \sec \theta$, $y = e^r \tan \theta$ prove $\frac{\partial(x,y)}{\partial(r,\theta)} \frac{\partial(r,\theta)}{\partial(x,y)} = 1$. (5)

- (b) Find the volume of the greatest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. (6)

UNIT III

15. (a) Evaluate $\int_0^3 \int_1^{\sqrt{4-y}} (x+y) dx dy$ by changing the order of integration.

- (b) Change the order of integration in

$$I = \int_{-a}^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) dx dy.$$

Or